Siratul Islam

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Education

Shahjalal University of Science and Technology — **Aug. 2023 – Aug 2027**
Bachelor of Science (Hons) in Physics — *Sylhet, Bangladesh*

- Relevant Coursework: Basic Electronics, Digital Electronics (upcoming), Programming with C (Theory+Lab, 4.00), Physics Lab: CRO/DSO calibration, Zener/Si diode characteristics, LCR series/parallel, full-wave bridge rectifier

Experience

Linux Foundation - Zephyr RTOS Project — **Remote**
Biometrics Subsystem Maintainer & Auth/FIDO2 Subsystem Maintainer (RFC) — *Feb. 2026 – Present*

- Author and Maintainer of Zephyr's biometrics subsystem (PR #100139), the first biometric subsystem in an RTOS
- Implemented reference drivers for ZFM-x0 & GT-5x optical fingerprint sensors
- Architecting the new FIDO2/CTAP2 authentication subsystem
- Establish technical standards and review all community contributions to the biometrics subsystem as assignee / reviewer
- Technologies: Zephyr RTOS, Embedded C, Subsystem Architecture, Device Drivers, Device Tree, Git

Contributor (Triage) — *Jun. 2025 – Present*

- Zephyr Contributor with Triage access and 20+ merged PRs across ARM, RISC-V, and Xtensa platforms
- Develop device drivers and board support packages serving embedded developers globally
- Review community contributions and provide technical guidance on driver architecture and device tree bindings
- Maintain official documentation for 9+ board support packages in Zephyr upstream

Linux Foundation - The Linux Kernel — **Remote**
Driver Maintainer, Industrial I/O (IIO) Subsystem — *Mar. 2026 – Present*

- **Maintained Drivers:** ST VL53L1X (Time-of-Flight sensor)
- Authored the driver and Device Tree bindings and used latest Kernel APIs
- Manage version control via lore.kernel.org, and collaborate with subsystem maintainers

Shahjalal University of Science and Technology — **Jun 2025 – Present**
Research Assistant - Department of Electrical and Electronic Engineering — *Sylhet, Bangladesh*

- Developing IoT-enabled smart control systems and radar-based occupancy detection for government energy optimization
- Building RFID-based attendance management systems for rural educational institutions in Bangladesh
- Building STM32F4 control system, implementing ROS2 nodes, developing OpenCV vision algorithms

Hackules Inc. — **Jun. 2024 – Jun. 2025**
Software Engineer — *Remote*

- Led full-stack/frontend development for educational platforms (Opedemy, Teachers Today) serving 1000+ active users

Selected Projects

Autonomous Vehicle Navigation System | *ROS2, C++, Jetson Orin, STM32F4* — **ongoing research**

- EEE department-funded project: ROS2-based autonomous navigation with LiDAR fusion and OpenCV computer vision
- Developed STM32F4 vehicle control system for real-time actuator management and motor control with sensor fusion

STM32 BME280 HAL Driver | *C, STM32, I2C, Bare-Metal* — **source code**

- Developed a custom Hardware Abstraction Layer (HAL) driver in C for the Bosch BME280 environmental sensor
- Implemented reliable I2C communication protocols to parse temperature, humidity, and pressure telemetry

TM1637 Linux Kernel Module | *C, Makefile, Device Tree* — **source code**

- Developed an out-of-tree (OOT) Linux loadable kernel driver for the TM1637 7-segment display
- Implemented sysfs interfaces and hardware communication layers for user-space interaction and brightness control

Technical Skills

Specializations : Zephyr RTOS, Linux Kernel, FIDO2/CTAP2, Embedded Biometrics, IoT Security & Hardware Auth
Embedded Systems: Zephyr RTOS, FreeRTOS, Linux Kernel, STM32 HAL, ESP-IDF, Embedded Linux, ROS2
Platforms & Protocols : STM32, ESP32, nRF52, Raspberry Pi, MQTT, BLE, Wi-Fi, LoRa, USB, NFC, I2C, SPI, UART
Development Tools: Embedded C/C++, Rust, STM32CubeIDE, CMake, PlatformIO, OpenOCD, KiCAD, Git
Full-Stack Web: C#, TypeScript, .NET, SvelteKit, Next.js, PostgreSQL, Docker, Podman

Publications

1. S. Islam, M. Rasedujjaman, "BASE: A Multi-Modal Biometric Architecture for Resource-Constrained RTOS," *Elsevier Internet of Things*, under review, 2026.